

COMP3314: Assignment 3

Image Classification Challenge

Team Members:

Yip Wing Long	3036389153
Chung Shing Hei	3036216760
Lam Chiu Long	3036222355

Date: April 11, 2026

1. Dataset Analysis

1.1 Statistical Overview

The dataset contains 50,000 training samples and 10,000 test samples, divided into 10 categories. The categories are mostly balanced, with standard deviation $\sigma = 35.45$, $\approx 0.7\%$ of the average. Unique Image Shapes found in sample $\{(32, 32, 3)\}$ indicate that the image samples have 3 channels and a 32x32 dimension.

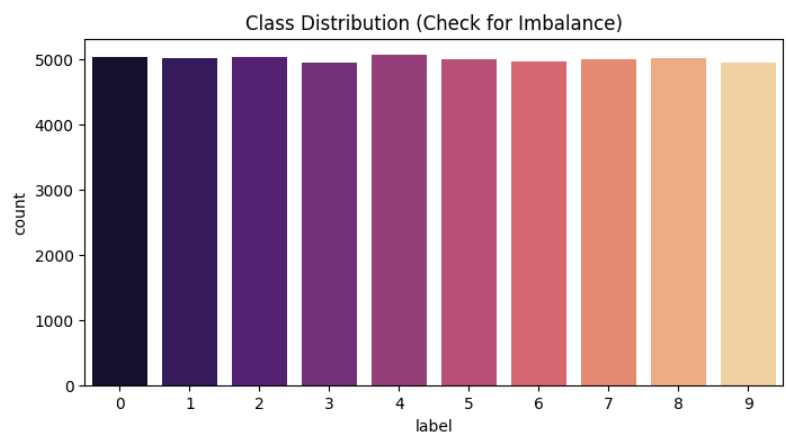
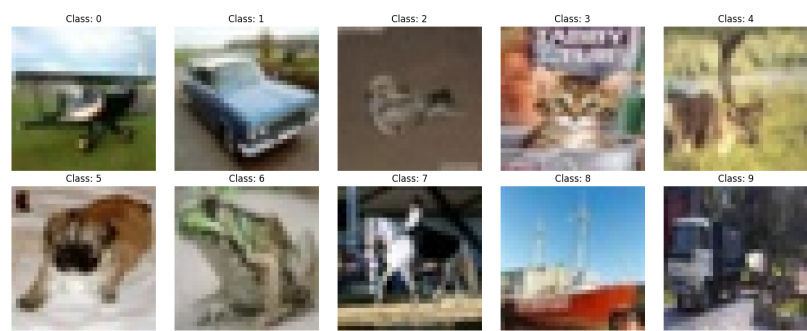


Figure 1: Class Distribution Count

0	1	2	3	4	5	6	7	8	9
5027	5008	5026	4946	5061	5005	4962	5006	5012	4947

Table 1: Number of Samples in Each Class

1.2 Category Visualization



The categories can be classified as:

0	1	2	3	4	5	6	7	8	9
Plane	Car	Bird	Cat	Deer	Dog	Frog	Horse	Ship	Truck

2. Classifier Exploration

SVM

```
classifier_chain = make_pipeline(  
    StandardScaler(copy=False),  
    PCA(n_components=0.85, random_state=42, copy=False),  
    SVC(kernel='rbf', C=10, class_weight='balanced', cache_size=100, random_state=42)  
)
```

Random Forest

```
classifier_chain = make_pipeline(  
    StandardScaler(copy=False),  
    PCA(n_components=0.85, random_state=42, copy=False),  
    RandomForestClassifier(n_estimators=200, max_depth=20, min_samples_split=5,  
                           class_weight='balanced', random_state=42, n_jobs=-1)  
)
```

LBGM

```
classifier_chain = make_pipeline(  
    StandardScaler(copy=False),  
    PCA(n_components=0.85, random_state=42, copy=False),  
    lgb.LGBMClassifier(**lgbm_params)  
)
```

Model	Accuracy	Key Observations
SVM	0.77 (0.765)	Performs better on large datasets
Random Forest	0.49 (on 50k images)	Suggesting it is not suitable for this classification task
LGBM	0.70	Performs similarly on datasets with 50k & 200k images

Detailed Observation:

Random Forest: The low accuracy of Random Forest suggested it wasn't a suitable model for this image classification task, so it was omitted from the final submission and future training (to save time).

Light Gradient Boosting Machine: LGBM performed well but showed signs of plateauing. Because decision trees perform axis-aligned splits, they can struggle with the highly correlated, high-dimensional feature space produced by image descriptors unless the tree depth is significantly increased, which risks overfitting.

(LGBM is a variation of Decision Tree. Since it is not in the slides, consider Random Forest as the second required model if LGBM is not allowed)

Support Vector Machine: The model achieved the highest accuracy, and therefore could be considered a reliable model for the task.

However, SVM performed poorly on animal classes, suggesting it is weaker at fine-grained category recognition compared to deep learning methods (CNNs/ResNet) or better featuring techniques may be needed.

3. Final Solution Description

Our final solution is a multi-stage pipeline that emphasizes high-quality feature extraction and rigorous validation.

3.1 Data Augmentation and Leakage Prevention

To handle the scale of the dataset without overwhelming system memory, the pipeline utilizes a procedural caching methodology:

- **Dynamic Caching:** Raw images and augmented variants are flushed directly to .npy arrays on disk (data).
- **Selected Augmentations:** To increase model robustness and generalization, the training set is expanded 4x using spatial transformations:
 - Baseline (Original)
 - Horizontal Flip (flip)
 - Spatial Shift Right by 4 pixels (shift_4)
 - Spatial Shift Up by 4 pixels (shift_up_4)

3.2 Feature Engineering (The Feature Union)

The pipeline extracts a densely concatenated matrix of handcrafted computer vision features. These features are dynamically generated and cached per augmentation:

- **Histogram of Oriented Gradients (HOG):** The most critical feature set. Multiple HOG resolutions are stacked to capture both fine details (best for animals) and macro structural shapes. Configurations include variations in orientations (9 and 18) and cell sizes (4x4 up to 10x10).
- **Local Binary Patterns (LBP):** Used to capture texture variations using two different configurations (LBP_1_8 and LBP_2_16).
- **Color Features:** A 3D Color Histogram generated in HSV space provides illumination-invariant color tracking, assisted by flattened raw color arrays (raw_color).

3.3 Strict Data Leakage Prevention

A critical methodological highlight is how the train/test splits are handled to prevent **augmentation leakage**:

- **Pre-Augmentation Splitting:** The dataset is split into training and validation indices (80/20) strictly on the *base* (core) image indices first.
- **Controlled Expansion:** Only the indices assigned to the training split are expanded to include their augmented variants.
- **Pure Validation:** The model is evaluated *only* on the 'Original' untouched images of the validation fold, giving a true representation of how it will perform on unseen test data.

3.4 The Classification Pipeline

The predictive modeling follows a clean, sequential scikit-learn Pipeline utilizing Intel's sklearnex extension to drastically speed up mathematical operations:

- **Standardization (StandardScaler):** Normalizes the widely varying scales of the concatenated HOG, LBP, and Color Histogram datasets.
- **Dimensionality Reduction:** Principal Component Analysis (PCA) is applied to retain **85% of the variance**. This drastically reduces the massive dimensionality of the concatenated features, filtering out noise and speeding up the SVM.
- **Support Vector Machine (SVC):** The final classifier is a Support Vector Classifier utilizing a Radial Basis Function (rbf) kernel. It uses $C=10$ (allowing a tighter fit to complex decision boundaries) and `class_weight='balanced'` to prevent bias toward majority classes.

(Note: While LGBM was also explored as an alternative gradient boosting option, the final pipeline structure currently defaults to the SVM model.)

4. Conclusion

By combining traditional computer vision descriptors with a robustly tuned SVM, we achieved a validation accuracy of **76.50%**. This result comfortably exceeds the 70% threshold for full marks. Our approach demonstrates that with careful feature engineering and a strict "no-leakage" validation strategy, classical machine learning remains a powerful tool for image classification tasks.

Appendix

Analysis of SVM

- Image set: 200k images, training set: 160k images, validation set: 10k original images
- The final CSV is predicted by a model trained using all 200k images

Running Diagnostics...

	precision	recall	f1-score	support
0	0.7900	0.8332	0.8110	1025
1	0.8637	0.8765	0.8700	1012
2	0.6863	0.7014	0.6938	1045
3	0.6081	0.5879	0.5979	995
4	0.7081	0.7074	0.7078	1022
5	0.6569	0.6471	0.6519	935
6	0.8155	0.8390	0.8270	1006
7	0.8174	0.7853	0.8010	992
8	0.8459	0.8425	0.8442	984
9	0.8519	0.8242	0.8378	984
accuracy			0.7650	10000
macro avg	0.7644	0.7644	0.7643	10000
weighted avg	0.7646	0.7650	0.7646	10000

Sub-Train Accuracy (Checking Overfitting): 0.0907

